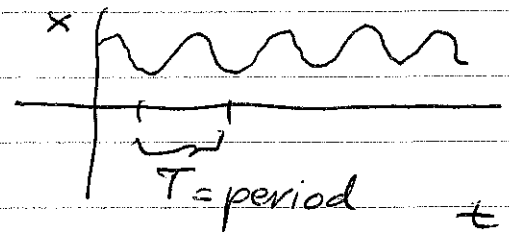
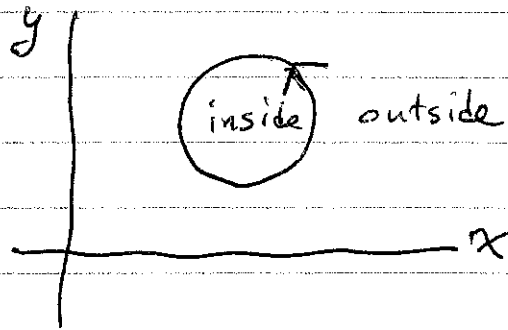


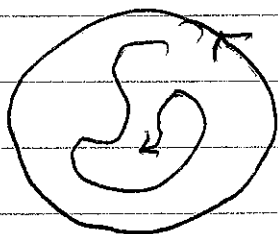
Limit cycle: isolated periodic orbit;
trajectory forms closed loop in phase space



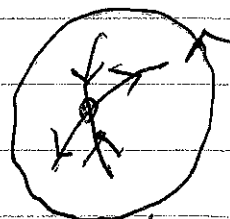
Note: in the phase plane there is an "inside" & "outside" and trajectories can't cross between.

What can trajectories do on the inside?
Can we say anything about possible fixed pts. on the inside of the limit cycle.

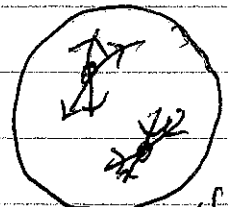
Turns out - none of the following is possible:



no fixed
pts.



exactly
one fixed
pt. of
saddle type

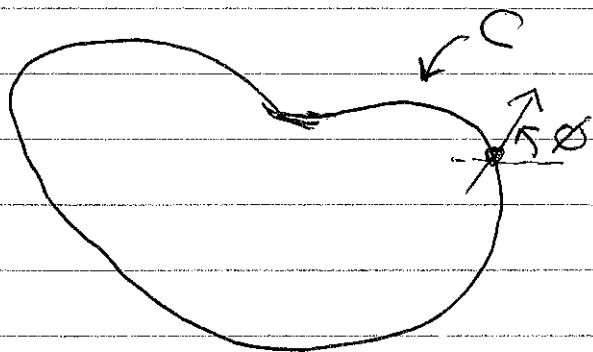


exactly
two fixed
pts - one
a stable
node; one an
unstable node

In $\mathbb{R}^2$, we can use "Poincaré Index" to rule out certain configurations of fixed pts inside a limit cycle.

Let $C =$ any simple closed curve that doesn't intersect any fixed pts. of the vector field

$$\left. \begin{array}{l} \dot{x} = f(x, y) \\ \dot{y} = g(x, y) \end{array} \right\} (f, g) \text{ is smooth vector field in } \mathbb{R}^2$$



$\phi =$ angle of vector field

$$\tan \phi = \frac{g(x, y)}{f(x, y)}$$

$I_C =$ "index of C " = net $\#$ of CCW rotations of ϕ as we go around C in CCW direction

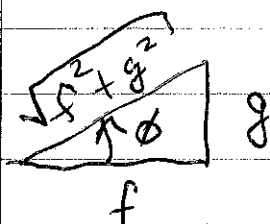
$$I_C = \frac{1}{2\pi} [\phi] = \frac{1}{2\pi} \oint_C d\phi$$

Lecture 9 p.3

If C was parameterized by θ so that $(x(\theta), y(\theta))$ defines a pt. on C with $\theta \in [0, 2\pi)$, $(x(0), y(0)) = (x(2\pi), y(2\pi))$ then we could compute I_C as an integral

$$\tan \phi(\theta) = \frac{g(x(\theta), y(\theta))}{f(x(\theta), y(\theta))}$$

$$\Rightarrow \sec^2 \phi(\theta) \frac{d\phi}{d\theta} = \frac{g'(\theta)f(\theta) - g(\theta)f'(\theta)}{f^2(\theta)}$$

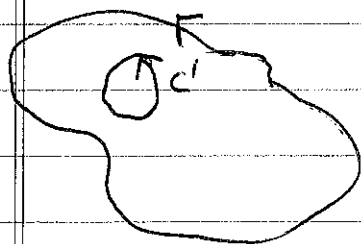


$$\frac{1}{\sec^2 \phi} = \cos^2 \phi = \frac{f^2}{f^2 + g^2}$$

$$I_C = \frac{1}{2\pi} \oint_C d\phi = \frac{1}{2\pi} \int_0^{2\pi} \frac{d\phi}{d\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} \frac{g'(\theta)f(\theta) - g(\theta)f'(\theta)}{f^2(\theta) + g^2(\theta)} d\theta$$

Important property of I_C : If C' is a

closed curve obtained by a continuous deformation of C , which doesn't cross any fixed-pts then

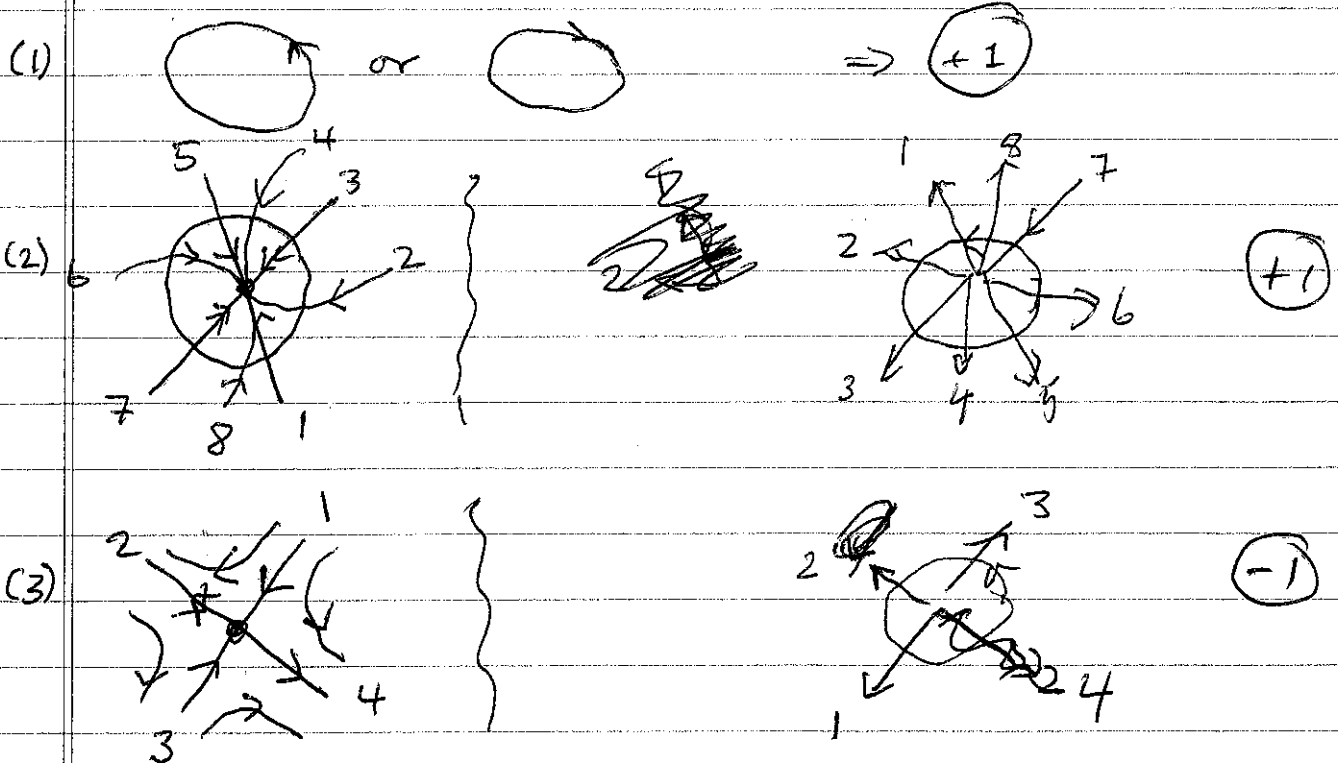


$$I_C = I_{C'}$$

This follows since I_C is an integer and it is a continuous function of the curve as C is deformed to C' . Hence it must be a constant ^{function} under the deformation.

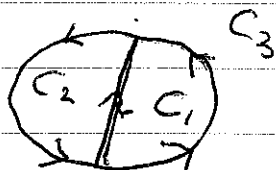
Additional properties of Poincaré Index:

- | | |
|--|--|
| (1) index of a periodic orbit = +1 | } index unchanged if we reverse time:
$(f, g) \rightarrow (-f, -g)$ |
| (2) index of a node, spiral, center = +1 | |
| (3) index of a saddle = -1 | |

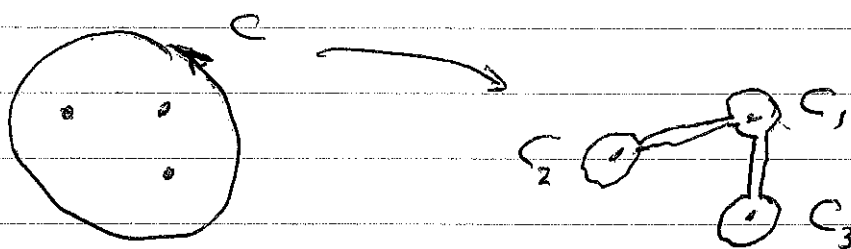


(4) index is additive

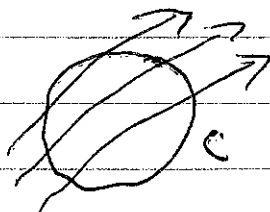
if $C_3 = C_1 + C_2$, then $I_{C_3} = I_{C_1} + I_{C_2}$



(5) index of a closed curve = sum of indices of fixed pts. inside it



(6) index of a closed curve with no fixed pts. inside it is zero



Corollary to above

If C is a limit cycle, then $I_C = \pm 1$

so sum of indices of fixed pts. inside = ± 1

If fixed pts. are isolated & hyperbolic then
 $\pm 1 = \sum \text{indices of fixed pts.}$ (need odd number)

Application: Competitive species model

$$\begin{aligned}\dot{N}_1 &= (r_1 - b_1 N_2) N_1 \\ \dot{N}_2 &= (r_2 - b_2 N_1) N_2\end{aligned}\quad r_1, r_2, b_1, b_2 > 0$$

↖ growth rate of species 2 diminished by presence of species 1.

non-dimensionalize $N_1, N_2, t \Rightarrow$ 4 parameters decreased to 1 parameter

$$\frac{dn_1}{d\tau} = n_1 - n_1 n_2 = n_1 (1 - n_2)$$

$$\frac{dn_2}{d\tau} = \rho n_2 - n_1 n_2 = n_2 (\rho - n_1)$$

$$t = \tau / r_1$$

~~$$N_1 = \frac{r_1}{b_2} n_1$$~~

$$N_1 = \frac{r_1}{b_2} n_1$$

$$N_2 = \frac{r_2}{b_1} n_2$$

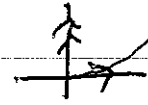
$$\rho = \frac{r_2}{r_1}, \text{ can assume } \rho \geq 1 \text{ w.l.o.g.}$$

- ① equilibria $n_1 = n_2 = 0$ extinction state
 $n_2 = 1, n_1 = \rho$ coexistence state

② linear stability: $DF = \begin{bmatrix} 1 - n_2 & -n_1 \\ -n_2 & \rho - n_1 \end{bmatrix}$

$$DF(0,0) = \begin{bmatrix} 1 & 0 \\ 0 & \rho \end{bmatrix}$$

unstable node



$$DF(p,1) = \begin{bmatrix} 0 & -\rho \\ -1 & 0 \end{bmatrix}$$

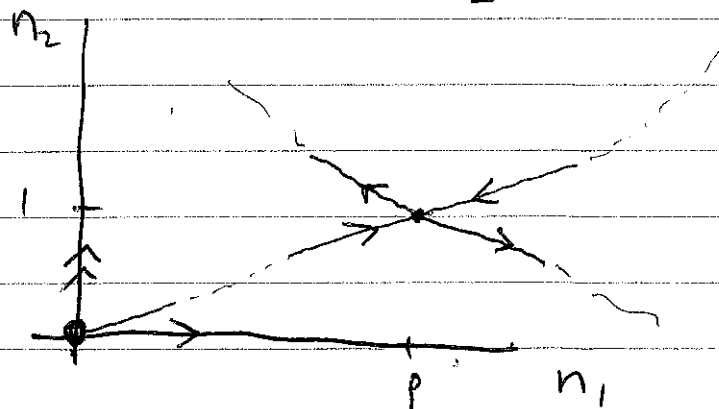
$$\left. \begin{array}{l} \tau = 0 \\ \Delta = -\rho \end{array} \right\} \begin{array}{l} \text{saddle} \\ \lambda = \pm \sqrt{\rho} \end{array}$$

$$\lambda = +\sqrt{\rho} \quad \begin{bmatrix} -\sqrt{\rho} & -\rho \\ -1 & -\sqrt{\rho} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -\sqrt{\rho} v_1 - \rho v_2 \\ -v_1 - \sqrt{\rho} v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$V = \begin{bmatrix} \sqrt{\rho} \\ -1 \end{bmatrix}$$

$$\lambda = -\sqrt{\rho} \quad \begin{bmatrix} \sqrt{\rho} & -\rho \\ -1 & \sqrt{\rho} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \sqrt{\rho} v_1 - \rho v_2 \\ -v_1 + \sqrt{\rho} v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$V = \begin{bmatrix} \sqrt{\rho} \\ 1 \end{bmatrix}$$



limit cycle? No
must enclose
one fixed
pt.
can't be
saddle.
can't cross
axes